NAME: SARANYA VIMALANATHAN **ROLL NO: 240701620**

USER INTERFACE AND DESIGN

Experiment – 3(a)

Create a prototype with familiar and unfamiliar navigation elements. Evaluate ease of use with different user groups using proto.io

Familiar Navigation

AIM:

The aim is to develop a prototype incorporating familiar navigation elements and assess usability among diverse user groups using Proto.io.

PROCEDURE:

1. Sign up or log in to proto.io and create a new mobile project.
2. Design screens using familiar UI components: top navigation bar, bottom tab bar, search bar, and labeled buttons.
3. Link screens using tap/click interactions and standard forward navigation transitions.
4. Share the prototype and gather feedback from user groups on ease of use.
5. Analyse how users navigate and identify pain points or positive responses.

OVERVIEW OF FAMILIAR NAVIGATION:

Familiar navigation relies on conventions users already know — tap-to-navigate buttons, bottom tab bars, search fields, and product listing cards. Users intuitively understand what each element does without needing instructions. This reduces cognitive load and speeds up task completion, since users apply mental models built from previous app experiences. The prototype below demonstrates a shopping app built entirely on these familiar patterns.

Group 1: Home Screen & Category Navigation

The home screen (f_1) presents a clean search bar at the top, two large labeled action buttons — 'View Sales' (navy) and 'View Cart' (green) — and a standard bottom navigation bar with icons for Home, Orders, Cart, and More. This layout mirrors patterns from popular e-commerce apps, making it immediately recognisable. The categories screen (f_2) extends this by displaying four category icons (Male, Female, Electronics, Plants) in a 2×2 grid, supported by the same familiar search bar and tab bar.

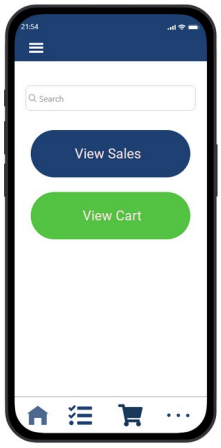

Fig 1: Home Screen

Fig 2: Category Screen

Group 2: Product Detail Screens

The product detail screens (f_3 to f_6) follow a standard e-commerce layout: a large product image at the top, availability status, product title, and a description below. A floating '+' button (blue circular FAB) allows users to quickly add items to the cart — a pattern widely used in Android apps. A right-arrow indicator on the image allows swiping to view more photos. Products shown include a Men's Formal Coat (not available), Black Sandals (available), a Desktop PC Bundle (not available), and a Natural Indoor Plant (available). The consistent structure across all product screens reduces the learning curve.

Fig 3: Men's Coat

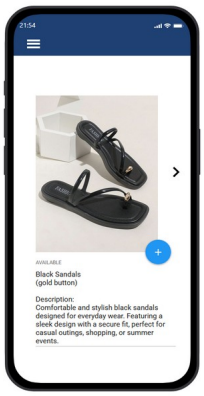

Fig 4: Black Sandals

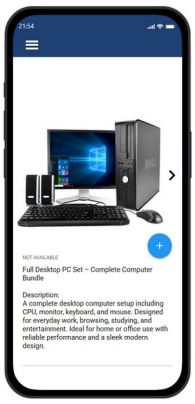

Fig 5: PC Bundle

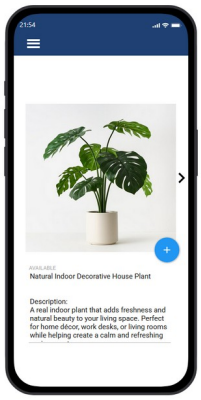

Fig 6: Indoor Plant

Group 3: Cart & Add-to-Cart Confirmation

The cart screen (f_7) displays added items as list rows with a thumbnail image and product name, again using the familiar search bar and bottom navigation. The add-to-cart confirmation screens (f_8 and f_9) show a large blue checkmark with 'Added to Cart' text — a universally understood success feedback pattern. Users immediately know their action was successful without needing to read instructions. This group of screens relies entirely on click/tap interactions that users expect from everyday shopping apps.

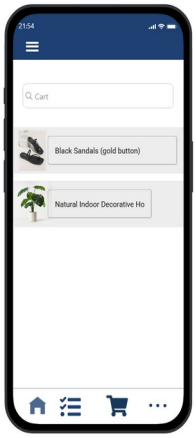

Fig 7: Cart View

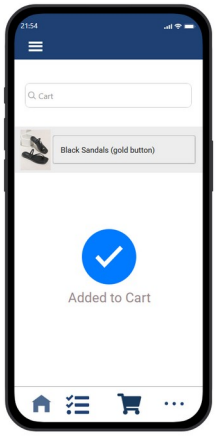

Fig 8: Sandals Added

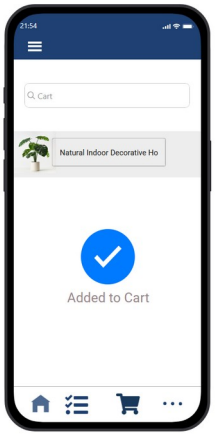

Fig 9: Plant Added

ANALYSIS:

- Users navigated the app quickly with minimal confusion due to the use of standard tap-to-proceed buttons and predictable screen layouts.
- The bottom tab bar and search bar served as anchor points, reducing disorientation even when deeper into the app.
- The floating action button ('+') for adding to cart was recognised by most users across all age groups.

- Success feedback (checkmark + 'Added to Cart') was immediately understood, confirming task completion without ambiguity.
- Familiar navigation lowers the learning curve and improves task efficiency, particularly for first-time users.

Unfamiliar Navigation

AIM:

The aim is to develop a prototype incorporating novel and unfamiliar navigation elements and assess the usability challenges they create across diverse user groups using Proto.io.

PROCEDURE:

6. Sign up or log in to proto.io and create a new mobile project.
7. Introduce non-standard navigation elements such as swipe-to-navigate gestures, unlabelled icon navigation, and inline directional arrows replacing buttons.
8. Set up interactions using gesture-based triggers (swipe left/right) for screen transitions.
9. Share the prototype with users from different age groups and collect task-completion data.
10. Document where users struggled, hesitated, or failed to navigate, and compare with familiar navigation results.

OVERVIEW OF UNFAMILIAR NAVIGATION:

Unfamiliar navigation introduces interaction patterns that deviate from standard conventions. Instead of tapping clearly labelled buttons, users are expected to swipe between screens, interpret unlabelled directional arrows (< >) placed beneath category icons, and deduce navigation pathways from contextual cues alone. These patterns require users to explore and discover functionality rather than relying on prior mental models. While this can create a more immersive experience for tech-savvy users, it significantly raises the difficulty level for older users, first-time users, and those less experienced with gesture-based interfaces.

Group 1: Swipe-Based Home Navigation

The modified home screen (u1) retains the visual layout of the familiar version — search bar, 'View Sales' and 'View Cart' buttons, and bottom navigation — but introduces a 'Swipe to Next page >' cue at the bottom of the content area. Rather than a clearly tappable navigation element, users must perform a horizontal swipe gesture to proceed. This is an unfamiliar pattern for users accustomed to tapping buttons. Many users, particularly older demographics, either missed the cue or attempted to tap the arrow instead of swiping, leading to navigation failure.

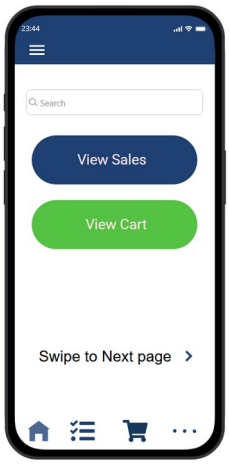

Fig 10: Unfamiliar Home – Swipe to Navigate

Group 2: Unlabelled Arrow-Based Category Navigation

The categories screen (u2) replaces the standard tap-to-enter grid navigation with a set of directional arrows (< >) placed below each category icon, paired with small text labels ('Male', 'Female', 'Plants', 'Electronics'). Users are expected to use these arrows to scroll or cycle through items within each

category rather than simply tapping the icon. This pattern is non-intuitive: the left and right arrows suggest a carousel or scroll interaction, but their purpose and behaviour are not immediately clear. Users unfamiliar with this pattern frequently tapped the icons directly or ignored the arrows entirely, resulting in incomplete task flows.

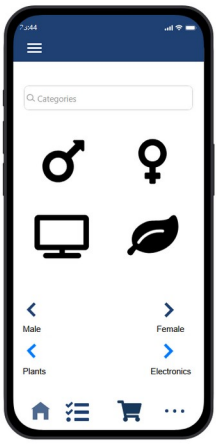

Fig 11: Unfamiliar Categories – Arrow Navigation

ANALYSIS:

- Users found swipe-based navigation difficult to discover; many attempted tapping the 'Swipe to Next page' text instead of performing a swipe gesture.
- The directional arrow (< >) pattern beneath category icons was confusing — users could not determine whether to click the arrows or the icons above them.
- Older and less tech-savvy users showed higher task abandonment rates compared to the familiar navigation prototype.
- Younger, tech-savvy users adapted more readily, though still experienced initial hesitation on first encounter with unfamiliar elements.
- Unfamiliar navigation significantly increased time-on-task and error rates, confirming that deviating from standard interaction patterns raises cognitive load.
- The comparison between 3(a) and 3(b) clearly demonstrates that familiar navigation conventions improve learnability, reduce friction, and produce a more positive user experience across all demographics.

Prototype Link:

Familiar : **https://pr.to/3QAVR7/**

Unfamiliar : **https://pr.to/NQ9XJO/**